



HWA CHONG INSTITUTION (COLLEGE SECTION)
2021 JC2 9744 H2 BIOLOGY
PRELIMINARY EXAMINATIONS PAPER 3 MARK SCHEME

QUESTION 1

- (a) Explain how the *ADA* mutation results in non-functional ADA enzyme in individuals with SCID.[4]
- ref to change in DNA sequence
 - ref to different mRNA transcribed
 - ref to different polypeptide formed
 - ref to loss of ADA enzyme active site
- (b) Explain what is meant by the term *recessive* in this context. [2]
- ref to idea of two copies of mutant alleles present
- (c)(i) With reference to **two** named T lymphocytes in the immune system, discuss how defective T lymphocytes increases the risk of infections. [4]
- ref to defective helper T cells and how their roles are affected
 - ref to cytotoxic T cells and how their roles are affected
- (ii) Suggest why an increased risk of infections can be life threatening for individuals with SCID. [1]
- ref to any logical reason
- (d) Suggest why stem cells are useful for gene therapy treatment. [2]
- ref to ideas of long-term self-renewal and differentiation
- (e) With reference to Fig. 1.2,
- (i) state the name of the enzyme that is used for the synthesis of RNA in viral genome replication.[1]
- RNA polymerase
- (ii) describe how the viral vector obtains its viral envelope. [3]
- ref to idea of budding with relevant details
- (f) Suggest **and** explain the functions of the *env* proteins located in the viral envelope in the gene therapy treatment. [2]
- ref to targeting and entry into specific host cells
- (g) Following the successful uptake of the viral vector into the target cell,
- (i) state the role of reverse transcriptase. [1]
- ref to idea of DNA synthesis
- (ii) suggest possible roles of the LTR that is flanking the therapeutic human gene. [2]
- ref to idea of expression of gene
- (h) Suggest why it is important that the viral vector lacks the *gag*, *pol* and *env* genes as shown in Fig. 1.1. [2]
- ref to idea of making viral vectors safe for gene therapy

- (i) Explain how the use of mRNA in RT-qPCR allows for quantification of the level of expression of the therapeutic gene. [2]
- ref to idea of a higher quantity of mRNA implying a higher level of expression of gene
- (j) With reference to Fig 1.3, identify **and** explain which individual **A** or **B**, would more likely to be successfully treated by gene therapy. [3]
- A
 - ref to any relevant data from Fig. 1.3
- (k) Suggest why individuals with SCID who have been successfully treated with gene therapy may still have offspring with SCID. [1]
- ref to any logical reason

[Total: 30]

QUESTION 2

- (a) State two similarities between ATP synthesis in a cyanobacterium and within a chloroplast. [2]
- ref to both requiring pumping H^+ across a membrane
 - formation of electrochemical gradient
- (b) Suggest and explain why the rate of photosynthesis in algae that live within the cells of coral polyps is higher than that of free-living algae. [3]
- coral / polyp undergo respiration
 - releases carbon dioxide concentration, higher / increased / no longer limiting factor
 - increase photosynthesis
- (c) With reference to Table 3.1, suggest why lamps radiating mainly violet and blue light are expected to increase the growth of coral polyps more than lamps radiating light of all wavelengths. [3]
- ref to peak pigments absorption
 - ref to rate of photosynthesis of algae increasing
 - ref to coral growth increases with more algal photosynthesis
- (d) Explain how a reduced cyanobacteria population will result in death of corals. [2]
- less nitrogen fixation
 - less cellular process like DNA replication

[Total: 10]

QUESTION 3

- (a) Describe and explain the results of the study shown in Fig. 3.1. [4]
- genome size of eukaryotes is larger than prokaryotes and quote relevant data
 - protein coding gene number of eukaryotes is larger than and quote relevant data
 - reason for why genome size of eukaryotes > prokaryotes
 - reason for why number of protein coding genes of eukaryotes > prokaryotes
- (b) Suggest two advantages of using a boxplot. [2]
- able to handle large amount of data
 - quick visual summary of the spread of data
 - provide indication of data's symmetry / skewness
 - outliers easily identified
- (c) Using your knowledge of molecular techniques, outline how the researcher can carry out this investigation. [4]
- use of polymerase chain reaction (PCR) to selectively amplify the DNA extracts, then add restriction enzymes to cut the DNA extracts
 - running agarose gel electrophoresis to separate the DNA extracts base on molecular size
 - transferring of DNA in the gel onto a membrane and denatured dsDNA to obtain ssDNA, to allow for fluorescent-labelled DNA probe to hybridise to the ssDNA
 - fluorescent bands observed for eukaryotic DNA extract as positive control
 - fluorescent bands observed for prokaryotic DNA extracts indicates *Cytochrome c* gene is present

[Total: 10]

QUESTION 4

- (a) Discuss how variation in viral genomes arises and suggest how a bird influenza virus could infect a human and subsequently form a new strain of human influenza virus. [15]
- ref to altered antigenicity
 - ref to antigenic drift
 - ref to gradual accumulation of mutations
 - ref to different types of hemagglutinin / neuraminidase
 - ref to errors during synthesis of nucleic acids
 - ref to factors influencing the rate of mutation
 - ref to antigenic shift
 - ref to reassortment of a segmented virus's genome with another genome
 - ref to sudden change in the antigenicity
 - ref to mutant hemagglutinin on viral membrane binding to sialic acid-containing receptors on host cell during adsorption
 - ref to virus being taken in by receptor-mediated endocytosis
 - ref to synthesis of new viral proteins
 - ref to new strain of virus acquiring human host cell surface membrane / viral envelope
 - ref to co-infection
 - ref to RNA genomes being replicated together
 - ref to RNA segments from both viruses being randomly packaged in one progeny virus

- (b) Prokaryotes differ from eukaryotes in terms of cell structure, genome organisation and cellular processes.

Outline the mechanism of asexual reproduction by binary fission in a typical bacterial cell and compare this with the mechanism of mitosis in a eukaryotic cell. [10]

- ref to bacterial chromosome being attached to the cell surface membrane
- ref to semi-conservative DNA replication
- ref to bidirectional replication
- ref to daughter DNA molecule being attached to the cell surface membrane
- ref to cell elongation causing the two chromosomes to separate
- ref to assembly of the septum
- ref to cell wall materials / peptidoglycan added
- (similarity) ref to separation of chromosomes
- (difference) ref to presence / absence of nuclear division
- (difference) ref to presence / absence of DNA replication
- (difference) ref to circular / linear chromosomes
- (difference) re. to disintegration of nuclear envelope or lack of it
- (difference) ref to movement of centrioles to opposite poles or lack of it
- (difference) ref to condensation of chromosomes or the lack of it
- (difference) ref to attachment of microtubules to chromosomes or the lack of it

[Total: 25]

QUESTION 5

- (a) Among all nucleated body cells, only B and T cells lose DNA during their development and maturation.

Describe processes where DNA is lost during the development of B cells before and after exposure to antigens and discuss the significance of this loss of DNA to the immune response. [15]

- somatic recombination
- process in which recombination happens in the heavy and light chain genes in the V region of the antibody
- V, (D), J gene segments are present in multiple copies in B cells
- Different V, D, J in heavy chains (V,J in light chains) are selected randomly during V(D)J rearrangement
- by deletion of intervening sequence and re-ligation of remaining gene segments
- V(D)J recombinase
- different V regions during development, creating a large diversity of antibody molecules
- loss of V, D, J segments allow for combinatorial diversity
- junctional diversity, with specific mention to loss of DNA
- deduces number of genes / size of genome needed to code for large number of different antibody molecules
- ref to being able to bind to wide variety of pathogens
- class switching after exposure to antigen
- same V_H exon can be rearranged
- to associate with different C_H exons / constant region of heavy chain
- DNA recombination in the heavy chain of Ig molecules between switch regions

- where excision of intervening DNA segments and subsequent ligation
- can result in the generation of antibodies of different isotypes
- ref to ability for subsequent switching to occur with any downstream C_H coding sequence
- somatic hypermutation
- specific method of DNA loss
- diverse response to the same antigen, as antigen specificity is retained, yet different effector activity through the different combinations of C_H exons
- affinity maturation
- allows for specific response at different times in the course of an infection
- speed of secondary response

(b) Charles Darwin postulated that over long period of time descent with modification eventually led to the rich diversity of life we see today.

Describe the mechanism for descent with modification and explain how variation in genetic information enables scientists to reconstruct phylogenies. [10]

- pre-existing genetic variation in population
- natural selection
- possible selection pressures eg. environment changes: food sources, predator
- individuals having selective advantage for survival
- individuals with selective advantage have higher survival rate and will live to reproductive age / differential survival and reproduction
- able to interbreed to produce viable, fertile offspring
- over time, change in frequency of favourable alleles in the gene pool of the population
- multiple sequence alignment to align comparable sequences
- ref. to comparing homologous genes / DNA sequences in different organisms
- DNA sequences from closely related organisms are more similar to one another
- The number of differences in DNA sequences between two groups of organisms may reflect how much time has passed
- since the group diverged from a common ancestor
- advantages of using genetic code (universal / unambiguous / quantifiable etc.)
- genetic sequences change over many generations due to accumulation of mutations
- mutations in nucleotide sequences taking place at a (constant rate)
- differences in allele frequencies to compute a genetic distance between two populations

[Total: 25]